

Founding Engineer Take-Home — ProcIndex

What you're building

An AI receptionist for a small pilates studio called Solstice Pilates. The studio gets around 30 calls a day. People want to book a class, change a booking, ask about pricing or hours, complain about a charge, ask if a friend can drop in, ask if you do birthday parties, or call to say they're running late. The agent picks up. You decide what it handles and what gets handed off to a human.

It talks to two things:

- Google Calendar, where it checks availability and creates or changes events.
- Google Sheets, where it keeps a row per caller and logs what came up on each call.

Phase 1 is a text chat. Phase 2 makes it a voice agent on Vapi.

Stack

TypeScript or Python. Any LLM provider.

Phase 1: build the agent

We only care about correctness here. Speed, token cost, and UI polish don't matter.

Build the agent and put a basic chat UI on top so we can talk to it. UI gets no points. Don't waste time on it.

For setup, put a few events on the Calendar in advance so you can test conflicts and reschedules. Make a Google Sheet with a Contacts tab. Columns are up to you.

A conversation might go something like:

Caller: Is the 6pm Reformer class on Thursday open? Agent: That one's full. The 7pm Reformer has two spots, want that instead? Caller: Sure, 7pm works. Agent: Got it. Can I get your name and a number to reach you on? Caller: Sara, 415-555-0190. Agent: Booked you in. Anything else?

Phase 2: voice with Vapi

Wire the Phase 1 agent into Vapi. It should sound like a real person. If there's more than about 1.2 seconds between someone speaking and the agent replying, it sounds like a bot. Vapi has a lot of knobs. Tune them.

Same conversation, on a call:

Caller: Hey, can I book the 6pm Reformer Thursday? Agent: Let me check... that one's full. 7pm has space, want that? Caller: Yeah okay. Agent: Cool. Name and number? Caller: Sara, four one five five five five oh one ninety. Agent: Got it Sara, you're in for 7pm Thursday. Anything else?

Submission

- GitHub repo (add reviewers if private).
- A short video of the Phase 1 chat. Calendar and Sheet should match what's on screen.
- A short video of a real call with the Phase 2 Vapi assistant.